

AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 22: Conjugate Gradient Method

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- 1 **CG as Optimization Method**
- 2 CG and Krylov Subspace
- 3 Convergence Properties of CG

Krylov Subspace Algorithms

- Create a sequence of Krylov subspaces for $Ax = b$

$$\mathcal{K}_k = \{b, Ab, \dots, A^{k-1}b\}$$

and find an “optimal” solution x_k in $\mathcal{K}_k$ at k th step

- Only matrix-vector products involved
- For SPD matrices, the most famous method is *Conjugate Gradient (CG)* method discovered by Hestenes/Stiefel in 1952
 - Finds best solution $x_k \in \mathcal{K}_k$ in norm $\|x\|_A \equiv \sqrt{x^T A x}$
 - Only requires storing 4 vectors (instead of k vectors) due to three-term recurrence

Motivation of Conjugate Gradients

- If $A \in \mathbb{R}^{n \times n}$ is SPD, then quadratic function

$$\varphi(x) = \frac{1}{2}x^T Ax - x^T b$$

has unique minimum

- Negative gradient of this function is residual vector

$$-\nabla \varphi(x) = b - Ax = r$$

so minimum is obtained precisely when $Ax = b$

- Optimization methods have form

$$x_{k+1} = x_k + \alpha_k p_k$$

where p_k is *search direction* and α_k is *step length* chosen to minimize $\varphi(x_k + \alpha_k p_k)$

- Line search parameter is $\alpha_k = r_k^T p_k / p_k^T A p_k$
- In CG, p_k is chosen to be A-conjugate (or A-orthogonal) to previous search directions, i.e., $p_k^T A p_j = 0$ for $j < k$

Conjugate Gradient Method

Algorithm: Conjugate Gradient Method

$$x_0 = 0, r_0 = b, p_0 = r_0$$

for $k = 1, 2, 3, \dots$

$$\alpha_k = (r_{k-1}^T r_{k-1}) / (p_{k-1}^T A p_{k-1}) \quad \text{step length}$$

$$x_k = x_{k-1} + \alpha_k p_{k-1} \quad \text{approximate solution}$$

$$r_k = r_{k-1} - \alpha_k A p_{k-1} \quad \text{residual}$$

$$\beta_k = (r_k^T r_k) / (r_{k-1}^T r_{k-1}) \quad \text{improvement this step}$$

$$p_k = r_k + \beta_k p_{k-1} \quad \text{search direction}$$

- Only one matrix-vector product $A p_{k-1}$ per iteration
- Apart from matrix-vector product, #flops per iteration is $\mathcal{O}(n)$
- If A is sparse with constant number of nonzeros per row, $\mathcal{O}(n)$ operations per iteration
- CG can be viewed as minimization of quadratic function $\varphi(x) = \frac{1}{2}x^T A x - x^T b$ by modifying steepest descent

An Alternative Interpretation of CG

Algorithm: CG

$$x_0 = 0, r_0 = b, p_0 = r_0$$

for $k = 1, 2, 3, \dots$

$$\alpha_k = (r_{k-1}^\top r_{k-1}) / (p_{k-1}^\top A p_{k-1})$$

$$x_k = x_{k-1} + \alpha_k p_{k-1}$$

$$r_k = r_{k-1} - \alpha_k A p_{k-1}$$

$$\beta_k = (r_k^\top r_k) / (r_{k-1}^\top r_{k-1})$$

$$p_k = r_k + \beta_k p_{k-1}$$

Algorithm: A non-standard CG

$$x_0 = 0, r_0 = b, p_0 = r_0$$

for $k = 1, 2, 3, \dots$

$$\alpha_k = r_{k-1}^\top p_{k-1} / (p_{k-1}^\top A p_{k-1})$$

$$x_k = x_{k-1} + \alpha_k p_{k-1}$$

$$r_k = b - A x_k$$

$$\beta_k = -r_k^\top A p_{k-1} / (p_{k-1}^\top A p_{k-1})$$

$$p_k = r_k + \beta_k p_{k-1}$$

- The non-standard one is less efficient but easier to understand
- It is easy to see $r_k = r_{k-1} - \alpha_k A p_{k-1} = b - A x_k$
- We need to show:
 - α_k minimizes φ along search direction p_k
 - α_k and β_k are equivalent to those in standard CG
 - Minimizing φ along p_k also minimizes φ within Krylov subspace

Optimality of Step Length

- Select step length α_k over vector p_{k-1} to minimize $\varphi(x) = \frac{1}{2}x^T Ax - x^T b$
- Let $x_k = x_{k-1} + \alpha_k p_{k-1}$,

$$\begin{aligned}\varphi(\alpha_k) &= \frac{1}{2}(x_{k-1} + \alpha_k p_{k-1})^T A(x_{k-1} + \alpha_k p_{k-1}) - (x_{k-1} + \alpha_k p_{k-1})^T b \\&= \frac{1}{2}\alpha_k^2 p_{k-1}^T A p_{k-1} + \alpha_k p_{k-1}^T A x_{k-1} - \alpha_k p_{k-1}^T b + \text{constant} \\&= \frac{1}{2}\alpha_k^2 p_{k-1}^T A p_{k-1} - \alpha_k p_{k-1}^T r_{k-1} + \text{constant}\end{aligned}$$

- Therefore,

$$\frac{d\varphi}{d\alpha_k} = 0 \Rightarrow \alpha_k p_{k-1}^T A p_{k-1} - p_{k-1}^T r_{k-1} = 0 \Rightarrow \alpha_k = \frac{p_{k-1}^T r_{k-1}}{p_{k-1}^T A p_{k-1}}.$$

- In addition, $p_{k-1}^T r_{k-1} = r_{k-1}^T r_{k-1}$ because $p_{k-1} = r_{k-1} + \beta_{k-1} p_{k-2}$ and $r_{k-1}^T p_{k-2} = 0$ due to the following theorem.

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Krylov Subspace in Conjugate Gradient

Theorem (Theorem 38.1 in NLA p. 295)

If $r_{k-1} \neq 0$, spaces spanned by approximate solutions x_k , search directions p_k , and residuals r_k are all equal to Krylov subspaces

$$\begin{aligned}\mathcal{K}_k &= \langle x_1, x_2, \dots, x_k \rangle = \langle p_0, p_1, \dots, p_{k-1} \rangle \\ &= \langle r_0, r_1, \dots, r_{k-1} \rangle = \langle b, Ab, \dots, A^{k-1}b \rangle\end{aligned}$$

The residuals are orthogonal (i.e., $r_k^\top r_j = 0$ for $j < k$) and search directions are A -conjugate (i.e., $p_k^\top A p_j = 0$ for $j < k$).

This theorem implies that

$$\alpha_k = (r_{k-1}^\top r_{k-1}) / (p_{k-1}^\top A p_{k-1}) = r_{k-1}^\top p_{k-1} / (p_{k-1}^\top A p_{k-1})$$

and

$$\beta_k = \frac{r_k^\top r_k}{r_{k-1}^\top r_{k-1}} = \frac{r_k^\top (r_{k-1} - \alpha_k A p_{k-1})}{r_{k-1}^\top r_{k-1}} = -\frac{r_k^\top A p_{k-1}}{p_{k-1}^\top A p_{k-1}}.$$

Proof of Properties of CG

Prove based on notation of standard CG.

- Proof of equality of subspaces by simple induction.
- To prove $r_k^\top r_j = 0$, note that $r_k = r_{k-1} - \alpha_k A p_{k-1}$ and $(A p_{k-1})^\top = p_{k-1}^\top A$, so

$$r_k^\top r_j = (r_{k-1} - \alpha_k A p_{k-1})^\top r_j = r_{k-1}^\top r_j - \alpha_k p_{k-1}^\top A r_j.$$

- If $j < k - 1$, then both terms on right are zero by induction.
- If $j = k - 1$, plug in $\alpha_k = (r_{k-1}^\top r_{k-1}) / (p_{k-1}^\top A p_{k-1})$

$$r_{k-1}^\top r_j - \alpha_k p_{k-1}^\top A r_j = r_{k-1}^\top r_{k-1} - r_{k-1}^\top r_{k-1} \frac{p_{k-1}^\top A r_{k-1}}{p_{k-1}^\top A p_{k-1}},$$

which is zero because

$$p_{k-1}^\top A p_{k-1} = p_{k-1}^\top A (r_{k-1} + \beta_{k-1} p_{k-2}) = p_{k-1}^\top A r_{k-1}$$

by induction hypothesis.

Proof Cont'd

- To prove $p_k^\top Ap_j = 0$, note that $p_k = r_k + \beta_k p_{k-1}$, so

$$p_k^\top Ap_j = r_k^\top Ap_j + \beta_k p_{k-1}^\top Ap_j.$$

- If $j < k - 1$, then both terms on right are zero by induction.
- If $j = k - 1$, plug in $\beta_k = (r_k^\top r_k)/(r_{k-1}^\top r_{k-1})$,

$$\begin{aligned} r_k^\top Ap_j + \beta_k p_{k-1}^\top Ap_j &= r_k^\top Ap_{k-1} + \frac{1}{\alpha_k} r_k^\top r_k \\ &= \frac{1}{\alpha_k} r_k^\top (r_k + \alpha_k Ap_{k-1}) \\ &= \frac{1}{\alpha_k} r_k^\top r_{k-1} \\ &= 0. \end{aligned}$$

Relationship with Lanczos Iteration

CG and Lanczos iteration are essentially the same process

- In CG, let b be right-hand side of $Ax = b$

$$\begin{aligned}\mathcal{K}_k &= \langle x_1, x_2, \dots, x_k \rangle = \langle p_0, p_1, \dots, p_{k-1} \rangle \\ &= \langle r_0, r_1, \dots, r_{k-1} \rangle = \langle b, Ab, \dots, A^{k-1}b \rangle\end{aligned}$$

- In Lanczos iteration for $A \in \mathbb{R}^{n \times n}$, starting from $q_1 = b/\|b\|$

$$AQ_k = Q_{k+1} \tilde{T}_k, \quad (1)$$

where $\tilde{T}_k$ is $(k+1) \times k$; Q_k is composed of orthonormal basis of $\mathcal{K}_k$

- If q_1 is a multiple of $r_0 = b$, then q_i will be proportional to r_{i-1}

- In (1), $\tilde{T}_k = \begin{bmatrix} \alpha_1 & \beta_1 & & \\ \beta_1 & \alpha_2 & \ddots & \\ & \ddots & \ddots & \beta_{k-1} \\ & & \beta_{k-1} & \alpha_k \\ & & & \beta_k \end{bmatrix}$

Alternative Derivation Based on Lanczos Iteration

- Let $x_k = Q_k y_k$. Then,

$$r_k = b - Ax_k = b - AQ_k y_k = b - Q_{k+1} \tilde{T}_k y_k$$

- Let $Q_k^\top r_k = Q_k^\top (b - Q_{k+1} \tilde{T}_k y_k) = 0$ (i.e., $r_k \perp \mathcal{K}_k$), we obtain

$$Q_k^\top Q_{k+1} \tilde{T}_k y_k = Q_k^\top b$$

where $Q_k^\top Q_{k+1} \tilde{T}_k = T_k$ and $Q_k^\top b = \beta e_1$ with $\beta = \|b\|$

- Hence,

$$T_k y_k = \beta e_1 \tag{2}$$

where $T_k = Q_k^\top A Q_k$ is tridiagonal, and is SPD if A is SPD

- It takes $\mathcal{O}(1)$ flops to update Cholesky factorization of T_k and then $\mathcal{O}(k)$ flops to solve (2). Resulting algorithm is equivalent to CG

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Termination in Exact Arithmetic

Theorem (Theorem 11.3.1 in MC p. 629)

If k_* is dimension of smallest invariant space that contains r_0 , then CG terminates in k_* steps in exact arithmetic.

- A subspace $\mathcal{S}$ is *invariant* w.r.t. A if for any $v \in \mathcal{S}$, $Av \in \mathcal{S}$

Proof

$r_0 = b$ can be written as a linear combination of k_* eigenvectors of A , $\{v_1, v_2, \dots, v_{k_*}\}$, so is $x_* = A^{-1}b$ (since A is diagonalizable).

At step $k = k_*$, $\dim(\mathcal{K}_{k_*}) = k_*$, and $\{v_1, v_2, \dots, v_{k_*}\}$ form a basis of $\mathcal{K}_{k_*}$, and hence $x_* \in \mathcal{K}_{k_*}$. If $x_* \in \mathcal{K}_k$ for $k < k_*$, $\dim(\mathcal{K}_k) = k < k_*$, then r_0 would have been contained in a lower-dimensional invariant space. Contradiction.

- If A has s distinct eigenvalues, CG converges in $\leq s$ iterations.
- With rounding errors, we may not get exact x_* after k_* iterations
- In addition, we may want to terminate sooner than k_* iterations

Optimality of Conjugate Gradients

Theorem (Theorem 38.2 in NLA p. 296)

If $r_{k-1} \neq 0$, then error $e_k = x_* - x_k$ is minimized in A -norm in $\mathcal{K}_k$.

Proof

Consider arbitrary point $x = x_k - \Delta x \in \mathcal{K}_k$ with error $e = x_* - x = e_k + \Delta x$. So

$$\begin{aligned}\|e\|_A^2 &= (e_k + \Delta x)^\top A (e_k + \Delta x) \\ &= e_k^\top A e_k + \Delta x^\top A \Delta x + 2e_k^\top A \Delta x,\end{aligned}$$

where $e_k^\top A \Delta x = r_k^\top \Delta x = 0$ because $r_k \perp \mathcal{K}_k$. Since A is SPD, $\|e\|_A^2 \geq \|e_k\|_A^2$ and equality holds iff $\Delta x = 0$.

- Because $\mathcal{K}_k$ grows monotonically, $\|e_k\|_A$ decreases monotonically
- Note: A -norm is defined as $\|x\|_A = \sqrt{x^\top A x}$, assuming A is SPD. It is different from weighted norm $\|x\|_W = \|Wx\|$

Convergence Rate with Rounding Errors

- If A has 2-norm condition number κ , error is bounded by

$$\frac{\|e_k\|_A}{\|e_0\|_A} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k$$

- Proof is based on analysis of matrix polynomials
 - CG minimizes $\|p_k(A)e_0\|_A$ at k th step, with $e_0 = x_*$, where p_k is degree- k polynomial $p_k(x) = 1 + c_1x + c_2x^2 + \dots + c_kx^k$
 - $\|e_k\|_A/\|e_0\|_A \leq \inf_{p_k} \max_{\lambda} |p_k(\lambda)|$, where λ are eigenvalues of A , which is further bounded using theory of orthogonal polynomials
- $2 \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} \right)^k \approx 2 \left(1 - \frac{2}{\sqrt{\kappa}} \right)^k$ for large κ , so CG takes up to $\mathcal{O}(\sqrt{\kappa})$ iterations
- In general, CG performs well with clustered eigenvalues